

Identifying the Drivers That Determine the Optimal Use of *Wolbachia* and Vaccination Programmes to Control Dengue

Abstract

Dengue is a growing public health threat, with over half the global population at risk. Dengue virus is mosquito-borne and consists of four antigenically distinct serotypes. Current strategies for reducing the burden of dengue include vaccination programmes and the release of *Wolbachia*-transinfected mosquitoes. Vaccination primarily reduces the dengue burden by protecting against severe disease. However, all currently licensed vaccines have serotype- and serostatus-specific heterogeneous efficacy profiles, producing an uneven distribution of risk of severe dengue by serotype and serostatus. Fixation of *Wolbachia*-carrying mosquitoes in the natural population acts to reduce the dengue burden by reducing the transmission intensity. Although, efficacy is dependent on environmental factors and there are concerns about long-term effectiveness. The optimal control strategy for dengue is likely to integrate both vaccination and *Wolbachia* release programmes. However, there are few published studies modelling this. This review discusses the current status of modelling these interventions. Drivers determining the optimal strategy are identified based on modelling attempts of both individual and combined interventions. This includes transmission intensity, one of the most influential drivers, as well as environmental factors, the circulating serotype, circulation of other arboviruses and socioeconomic factors. The review concludes with a proposal for a project investigating the impacts of routine vaccination with TAK-003 and *Wolbachia* release in Indonesia.

1 Introduction

Dengue is a mosquito-borne systemic viral infection (Paz-Bailey et al., 2024). While the majority of infections are asymptomatic (Asish et al., 2023), symptomatic manifestations range from mild febrile illness to severe dengue, characterised by severe bleeding, plasma leakage or organ impairment (WHO, 2024). Currently, dengue treatment is restricted to supportive management, as there are no clinically approved antiviral treatments (WHO, 2024). DENV strains are transmitted between human hosts by *Aedes (Stegomyia)* spp. vectors, primarily *Aedes aegypti* (Weaver and Reisen, 2010). Dengue is prevalent across tropical and sub-tropical regions (Bhatt et al., 2013; Liu-Helmersson et al., 2014) with 5.66 billion people, across 169 countries, estimated to be at risk based on environmental suitability (Lim et al., 2025). The global burden of dengue is rising, largely due to urbanisation, population movement and climate change (Wilder-Smith et al., 2017). Modelling based on environmental and socioeconomic factors predicted an increase in the at-risk global population to over 6.1 billion by 2080 (Messina et al., 2019). This increasing threat necessitates the development and implementation of safe and effective interventions.

2 Biological Background of Dengue Infection and Transmission

Dengue consists of four antigenically distinct serotypes (DENV-1 to DENV-4), each of which consists of numerous genetically distinct strains (Lee et al., 2025). Infection with one serotype is thought to induce lifelong homotypic immunity and temporary heterotypic immunity (Gibbons et al., 2007; Reich et al., 2013), although estimations of the duration of heterotypic immunity vary from 2 months to over 2 years (OhAinle et al., 2011; Reich et al., 2013; Sabin, 1952; Snow et al., 2014). A significantly elevated risk of severe dengue upon secondary infection, relative to primary infection, is well documented (Burke et al., 1988; Graham et al., 1999; Halstead, 1970; Sangkaew et al., 2021; Sangkawibha et al., 1984; Thein et al., 1997). This increase in risk has been largely attributed to antibody-dependent enhancement, where binding of the second serotype by weakly neutralising antibodies from the primary infection results in infection enhancement of FcγR bearing myeloid cells

(Goncalvez et al., 2007; Halstead, 2014; Wahala and De Silva, 2011). The proportion of hospital admissions for dengue due to tertiary or quaternary infection is extremely low (0.08-0.8%) (Gibbons et al., 2007). There is spatiotemporal variation in the predominating serotypes (Nisalak et al., 2003; San Martín et al., 2010; WHO, 2024) and serotype-specific heterogeneity in pathogenicity, although the differences reported vary (Burke et al., 1988; Graham et al., 1999; Halstead, 1970; Sangkaew et al., 2021).

3 Interventions

Traditional dengue control strategies have centred around the suppression of mosquito populations through mechanical, biological and chemical intervention (Achee et al., 2015; Araújo et al., 2015). These strategies have achieved short-term reductions in mosquito populations on local scales. However, these measures may be ineffective in the long-term, with evidence that mosquito populations increase back to pre-treatment levels after interventions conclude (Esu et al., 2010). Furthermore, there is evidence that large-scale insecticide releases would promote the evolution and spread of insecticide resistance, sparking ecological and efficacy concerns (Hassan et al., 2021; Luz et al., 2011; Moyes et al., 2017). Current research into potential control strategies focuses on the integration of *Wolbachia*-transinfected mosquito release and vaccination programmes (WHO, 2024).

3.1 Vaccination Programmes

Dengue vaccine development has proved particularly challenging, largely due to the immunological interactions between the four serotypes (Goncalvez et al., 2007; Izmirly et al., 2020). Efforts have been ongoing for over 60 years (Bellanti et al., 1966), with the first vaccine, CYD-TDV (Dengvaxia®), licensed in 2015 (Preto et al., 2021). Modelling predicts that effective vaccination programmes primarily reduce the burden of disease by lowering the individual risk of severe dengue in seropositive recipients, rather than reducing transmission (Flasche et al., 2016). However, vaccine-induced immunity may increase the risk of severe disease in seronegative recipients (Flasche et al.,

2019; Sridhar et al., 2018). It is hypothesised that vaccination of a seronegative individual mimics primary infection, placing the recipient at an elevated risk of severe dengue upon subsequent infection, due to a secondary-like immune response (Guy and Jackson, 2016). Therefore, the ideal vaccine would be equally and highly effective against all four serotypes (Rodriguez-Barraquer et al., 2014). However, all currently licensed vaccines have heterogeneous serotype- and serostatus-specific efficacy profiles (Table 1 and 2). There are two vaccines currently licensed for use, CYD-TDV (Dengvaxia®) and TAK-003 (Qdenga®), with one more having completed phase 3 clinical trials, TV-003/TV-005 (Butanan-DV), although the final data has not yet been released (Wilder-Smith et al., 2025).

Vaccine	Age group	Doses	Efficacy, %					References
			Overall	DENV-1	DENV-2	DENV-3	DENV-4	
CYD-TDV	2-17 years	3	60	51	34	75	77	Rosa et al. (2019)
TAK-003	4-16 years	2	73.3	69.8	95.1	48.9	51.0	Biswal et al. (2020)
TV-003/ TV-005	2-59 years	1	79.6	89.5	69.6	N/D	N/D	Kallás et al. (2024)

Table 1 | Comparison of overall and serotype-specific vaccine efficacies, %, against symptomatic virologically confirmed dengue (VCD) between CYD-TDV, TAK-003 and TV-003/TV-005. Adapted from Lee et al. (2025).

N/D = Not determined. These inconclusive results were due to insufficient DENV-3 and DENV-4 cases.

Vaccine	Age group	Doses	Efficacy, %		References
			Seropositive	Seronegative	
CYD-TDV	9-16 years	3	83.7	43.2	Villar et al. (2015)
TAK-003	4-16 years	2	76.1	66.2	Biswal et al. (2020)
TV-003/TV-005	2-59 years	1	89.2	73.6	Kallás et al. (2024)

Table 2 | Comparison of serostatus-specific vaccine efficacies, %, against symptomatic VCD between CYD-TDV, TAK-003 and TV-003/TV-005. Adapted from Lee et al. (2025).

The heterogeneous efficacy profiles lead to an uneven distribution of risk across individuals by serostatus and infecting serotype. Seronegative recipients of CYD-TDV may experience an increase in their risk of severe dengue particularly during a DENV-2 outbreak (Table 3) (Laydon et al., 2021; Lee et al., 2025). Seronegative recipients of TAK-003 may experience an increase in their risk of severe dengue particularly during a DENV-3 or DENV-4 outbreak (Table 4) (Cracknell Daniels et al., 2025).

Serotype	Seronegative		Seropositive	
	Efficacy, %	(95% CI)	Efficacy, %	(95%, CI)
DENV-1	15.4	(-23.3 to 45.0)	70.2	(57.4 to 80.1)
DENV-2	-12.5	(-42.2 to 26.8)	67.9	(49.9 to 82.0)
DENV-3	45.3	(-20.2 to 99.7)	77.5	(64.3 to 90.2)
DENV-4	51.2	(20.0 to 72.8)	89.9	(79.8 to 99.9)
All	21.9	(-5.5 to 42.6)	75.4	(68.3 to 81.6)

Table 3 | Efficacy, %, of CYD-TDV against symptomatic VCD, stratified by serotype and serostatus. Data from Yang et al. (2018). Abbreviation: CI – Confidence Interval

Serotype	Seronegative		Seropositive	
	Efficacy, %	(95% CI)	Efficacy, %	(95%, CI)
DENV-1	45.4	(26.1 to 59.7)	56.1	(44.6 to 65.2)
DENV-2	88.1	(78.6 to 93.3)	80.4	(73.1 to 85.7)
DENV-3	-15.5	(-108.2 to 35.9)	52.3	(36.7 to 64.0)
DENV-4	-105.6	(-628.7 to 42.0)	70.6	(39.9 to 85.6)
All	53.5	(41.6 to 62.9)	64.2	(58.4 to 69.2)

Table 4 | Efficacy, %, of TAK-003 against symptomatic VCD, stratified by serotype and serostatus. Data from Tricou et al. (2024). Note that the number of VCD cases was low for the placebo group of DENV-4.

Due to the potential increase in risk of severe dengue in seronegative recipients, WHO recommends implementation of CYD-TDV only with pre-vaccination serostatus testing (WHO, 2024). Conversely, for TAK-003, WHO does not recommend a pre-vaccination screening strategy in high-transmission settings as the public health impact would be substantially reduced and programmatic costs increased.

The potential discrepancy between individual-level risks and population-level benefits regarding blanket vaccination strategies (i.e. no pre-vaccination screening) raises ethical concerns. However, the risk to seronegative recipients is mitigated if the direct protection against some serotypes and any indirect effect of reducing transmission compensates for any increase in risk of symptomatic virologically confirmed dengue (VCD) or hospitalisation upon infection with other serotypes (Cracknell Daniels et al., 2025; Ferguson et al., 2016). Although modelling blanket vaccination with CYD-TDV predicted a small reduction in transmission intensity (Flasche et al., 2016), there is very limited research on the influence of TAK-003 on transmission intensity (El Hindi

et al., 2025). Nonetheless, blanket vaccination is predicted to provide an individual-level positive effect to seronegative recipients in high-transmission settings even under the assumption that vaccination with TAK-003 does not protect against infection; the high probability of multiple natural infections and protection against DENV-1 and DENV-2 outweigh the increase in risk associated with subsequent DENV-3 and DENV-4 infection (Cracknell Daniels et al., 2025).

Results from phase 3 trials reveal that TV-003/TV-005 also has heterogeneous serotype- and serostatus-specific efficacy profiles. Still, the overall and serostatus-specific efficacies are higher than those for CYD-TDV and TAK-003 (Table 1).

3.2 Release of *Wolbachia*-Transinfected Mosquitoes

More recently, strategies involving the release of *Wolbachia*-transinfected *Ae. aegypti* have been trialled for dengue control (Utarini et al., 2021). *Wolbachia pipientis* is a maternally-inherited endosymbiotic intracellular bacterium, naturally found in 66% of insect species (Dorigatti et al., 2018; Hilgenboecker et al., 2008; Utarini et al., 2021). This includes *Aedes albopictus*, a less significant dengue vector, but not *Ae. aegypti*, the main vector of dengue (Dorigatti et al., 2018; Gloria-Soria et al., 2018; Wright and Barr, 1980). The establishment of *Wolbachia* in the natural *Ae. aegypti* population acts to reduce the dengue burden by reducing transmission intensity (Ferguson et al., 2015). In a trial conducted in Yogyakarta, Indonesia, Utarini et al. (2021) found that introgression of wMel, a strain of *Wolbachia*, into intervention clusters was 77.1% effective for the prevention of symptomatic VCD, with similar efficacy against all four serotypes (Table 5).

Wolbachia strains developed for *Ae. aegypti* transinfection confer a reproductive advantage of cytoplasmic incompatibility; eggs from uninfected females and *Wolbachia*-carrying males are inviable (Ross et al., 2022). However, *Wolbachia* may also confer strain-specific fitness costs, such as delayed development (Ross et al., 2014), reduced survival of desiccated eggs (Ritchie et al., 2015), reduced lifespan (McMeniman et al., 2009) and reduced fecundity (Maciel-de-Freitas et al., 2024). This establishes a trade-off between the conferred reproductive advantage and fitness cost which,

along with the degree of vertical transmission (Fine, 1978), determines the frequency-dependent invasion threshold for *Wolbachia* establishment in the natural population (Dorigatti et al., 2018; Turelli and Barton, 2017).

The invasion of *Wolbachia*-transinfected mosquitoes into a wild population exhibits bistable dynamics. Transinfected mosquitoes must be released at a quantity such that the frequency-dependent invasion threshold is exceeded to ensure a high likelihood of fixation in the wild population. However, local fixation does not guarantee spatial spread. The size of the release area necessary to initiate spatial spread is dependent on the initial infection frequency, critical threshold frequency, host dispersal patterns and environmental characteristics (Dorigatti et al., 2018).

Serotype	Efficacy, %	(95% CI)
DENV-1	71.0	(18.2 to 89.7)
DENV-2	83.8	(72.1 to 90.6)
DENV-3	75.3	(4.9 to 93.6)
DENV-4	73.6	(57.9 to 83.5)
All	77.1	(65.3 to 84.9)

Table 5 | Efficacy, %, of wMel against symptomatic VCD, stratified by serotype. Data from Utarini et al. (2021).

4 Discussion: Drivers That Determine the Optimal Control Strategy

For this review, a driver is defined as an intrinsic property, underlying process or condition which influences the performance of an intervention and, therefore, which strategy is optimal for given set of conditions.

4.1 Transmission Intensity

Transmission intensity refers to the rate at which dengue spreads throughout the host population. In modelling, it is typically quantified by the force of infection (FOI), defined as the per capita rate of infection of susceptible individuals, or the reproduction number (R_0), defined as the average number of infections caused by a single infectious individual in a susceptible population (Cattarino et al., 2020). However, the average seroprevalence in 9-year-olds (SP9) is often employed as a proxy

(Cracknell Daniels et al., 2025; Flasche et al., 2016) as estimating FOI or R_0 can be challenging (Johansson et al., 2011).

Transmission intensity is one of the most influential drivers for determining the optimal control strategy. Modelling transmission intensity implicitly accounts for many compounding factors (Box 1). For vaccination strategies, accounting for transmission intensity is critical for minimising the risk to seronegative individuals. As transmission intensity increases, seroprevalence increases and the average age of first infection decreases (WHO, 2024).

Box 1: Factors influencing transmission intensity

- Entomological factors include: average daily biting rate, probability of vector-human and human-vector transmission per bite, extrinsic incubation period duration, vector mortality rate, vector competence and vector density per person (Lambrechts et al., 2011; Liu-Helmersson et al., 2014).
- Viral and immunological factors: The probability of human-vector transmission per bite is itself influenced by the level of viraemia: a mosquito is more likely to be infected when viraemia levels are higher (WHO, 2024). Additionally, antibody-dependent enhancement increases viral load (Lee et al., 2025).
- Climatic factors include temperature and precipitation (Bhatt et al., 2013). Dengue epidemic potential peaks at an average temperature of 29°C when the diurnal temperature range (DTR) is 0°C. This may be because shorter extrinsic incubation periods and higher biting frequencies are associated with higher mean temperatures. The effect of DTR is dependent on the mean temperature. When the mean temperature is close to 29°C, a smaller DTR increases the epidemic potential whereas larger DTRs increase the epidemic potential in cold to temperate or extremely hot settings (Liu-Helmersson et al., 2014). Precipitation may influence transmission intensity by creating breeding sites and stimulates egg hatching (Johansson et al., 2009).
- Socioeconomic factors: Densely populated regions are associated with high transmission (Bhatt et al., 2013); overcrowding creates a dense pool of hosts and man-made water storage containers provide breeding sites (Morrison et al., 2004).

To minimise risk, routine vaccination should target age groups with a high expected seroprevalence, with the optimal target age decreasing as transmission intensity increases (Cattarino et al., 2020). In high-transmission settings, blanket vaccination of school-age children could provide large public health benefits (WHO, 2024). Using a model calibrated with clinical trial data, Cracknell Daniels et al. (2025) estimated an average reduction in the burden of hospitalised dengue by 10-22% in moderate-to-high transmission settings ($SP_9 > 60\%$) over 10 years upon blanket vaccination with TAK-003 of children aged over 6 years. Additionally, as discussed in section 3.1, an individual-level net positive effect on seronegative recipients was predicted upon application of this strategy in high-transmission settings ($SP_9 > 60\%$), as recommended by WHO. Furthermore, in high-transmission settings individuals are highly likely to experience at least two natural infections. Considering this, the practical effect of vaccination can be thought of as a decrease in the age of experiencing a secondary-like infection (Flasche et al., 2016).

Implementing a blanket vaccination strategy in a low-transmission setting may increase the burden of dengue, due to the high proportion of seronegative recipients. All models compared by Flasche et al. (2016) predicted an increase in the number of hospitalisations upon routine vaccination of 9-year-olds using CYD-TDV with 80% coverage in very-low-transmission settings ($SP_9 = 10\%$). Modelling in low-transmission settings ($SP_9 = 30\%$) produced variable results on the number of hospitalisations, with four models predicting a reduction and four models predicting an increase. Modelling by Ferguson et al. (2016) predicted that targeting routine vaccination with CYD-TDV to an age group where seroprevalence exceeds $\sim 35\%$ avoids any negative population-level impact. This suggests that the routine vaccination of 9-year-olds with CYD-TDV in a low-transmission setting could have a net-negative population-level impact, in agreement with Flasche et al. (2016).

Screen-and-vaccinate programmes, where individuals are only vaccinated if they test seropositive, may offer a solution to this. Using the transmission model of Ferguson et al. (2016), 11 to 31% reductions in symptomatic dengue cases were predicted by Cattarino et al. (2020) upon the application of a pre-screening vaccination policy over the first 30 years. A policy coverage of 80% was assumed, along with 90% sensitivity and 95% specificity for seropositivity testing. Furthermore,

application in low-transmission settings had a higher predicted relative impact on dengue incidence than high-transmission settings. WHO (2024) has recommended screen-and-vaccinate programmes using CYD-TDV in low-transmission settings. However, subsequent low global demand of CYD-TDV, potentially, in part, due to operational hurdles associated with pre-vaccination screening, has led to the discontinuation of its manufacturing by Sanofi Pasteur (Cracknell Daniels et al., 2025).

It could be beneficial to consider how vaccination strategies could be combined with *Wolbachia* release, as this may result in a more effective strategy. Combining interventions may compensate for problems associated with the individual interventions. For vaccination, some individuals may not be eligible, and, in practice, specific age groups are targeted, potentially leaving other demographics more vulnerable. Logistical and financial constraints also apply to *Wolbachia* releases.

Wolbachia release reduces the burden of disease by decreasing transmission intensity (Ferguson et al., 2015), suggesting long-term integration of *Wolbachia* release and blanket vaccination may be incompatible, due to the potential negative impact of blanket vaccination in low-transmission settings. Despite this, modelling by Ndi (2020) predicted a 99% reduction in the number of dengue cases with a combined strategy, compared to 19% and 92% reductions for vaccination and *Wolbachia* only, respectively. However, a single serotype model was used, meaning the effects of secondary infections and potential increase in risk for seronegative recipients were not accounted for. Additionally, vaccine efficacy for seronegative recipients (due to the single serotype model set-up), was assumed to be 53.6%. Across both CYD-TDV and TAK-003, only the efficacy of TAK-003 against DENV-2 exceeds this for seronegative recipients. There is a need for multi-serotype transmission models testing combined blanket vaccination and *Wolbachia*-based strategies, accounting for the complex efficacy profile of the available vaccines.

Conversely, screen-and-vaccinate programmes may be more compatible with *Wolbachia* release, due to their effectiveness in low-transmission settings. Ndi et al. (2024) predicted that the combination of *Wolbachia* release and vaccination of seropositive individuals resulted in a higher reduction in secondary infections than *Wolbachia* release alone. However, the model did not explicitly

account for pre-vaccination screening. Test sensitivity and specificity were not modelled and, therefore, the effects of vaccinating of a small proportion of seronegative individuals were not considered. More research is needed into the influence of combining pre-vaccination screening strategies with *Wolbachia* release.

4.2 Spatial Heterogeneity in Transmission Intensity

Urbanisation and increasing population densities mean that areas of high transmission concentrate in a relatively small urban area. O'Reilly et al. (2019) estimated that 90% of cases occur in 15.26% of the land area. Hence, *Wolbachia* release, which scales with land area rather than population, may be more impactful and cost-effective in these areas (O'Reilly et al., 2019). Mapping the variations in transmission intensity can be used to pinpoint high priority areas for *Wolbachia* release (Cattarino et al., 2020; O'Reilly et al., 2019; Van Panhuis et al., 2015). This could potentially be combined with screen-and-vaccinate programmes in rural areas.

4.3 Serotype

The effectiveness of a vaccination programme for a particular area is dependent on the serotype-specific efficacy profile of the vaccine and the dominant circulating serotype. For example, vaccination programmes using TAK-003 in an area endemic to DENV-3 are likely to have lower effectiveness than CYD-TDV in reducing the burden of disease due to its lower overall and serostatus-specific efficacy against DENV-3 (Tables 1, 3 and 4).

4.4 Temperature

In addition to influencing transmission intensity independently of intervention, temperature also impacts the performance of *Wolbachia*. Heatwaves may reduce *Wolbachia* density, maternal transmission and expression of cytoplasmic incompatibility, although sensitivity is strain-specific

(Ross et al., 2017). While preliminary findings suggest that wMel exhibits an overall stronger inhibition of dengue (Sodeinde et al., 2025), it is also more sensitive to temperature cycling than wAlbB (Ross et al., 2017). Hence, diurnal cyclical temperatures influence the suitability of a *Wolbachia* strain for a particular location.

4.5 Prevalence of Other Arboviruses

Unlike vaccination, *Wolbachia* is effective against other arboviruses. Pinto et al. (2021) reported a reduction of 69% in dengue incidence, 56% in chikungunya incidence and 37% in Zika incidence upon wMel intervention in a quasi-experimental study in Brazil. While the estimated global case numbers of other arboviruses are much lower than dengue (Franklinos et al., 2019), *Wolbachia*-based strategies may still be particularly advantageous to areas prone to their outbreaks.

4.6 Socioeconomic Factors

Public health decisions usually operate within fixed budget constraints. Knerer et al. (2021) modelled how the optimal combination of *Wolbachia* and paediatric vaccination coverage, to minimise the number of dengue cases and disability-adjusted life years lost, varies by budget constraint. Under base-case budget constraints ($\leq$ \$368 million), the predicted optimal coverage was 50% *Wolbachia* with 49% vaccination coverage. In comparison, under the lowest budget constraint ($\leq$ \$251 million), the predicted optimal coverage was 54% *Wolbachia* with 10% vaccination coverage, demonstrating how smaller budgets necessitate trade-offs in resource allocation. However, the potential implementation of screen-and-vaccinate strategies was not assessed. While modelling by Coudeville et al. (2020) predicted that screen-and-vaccinate strategies are likely to be cost-effective in settings where $SP9 \leq 30\%$, its integration into an optimal strategy has not been assessed using a constrained optimisation approach.

Strong public confidence is essential to achieve high vaccine uptake (Wilder-Smith et al., 2025), so community engagement is a fundamental driver in the success of vaccination programmes. Following an announcement in November 2017 that CYD-TDV had previously unknown risks for seronegative recipients, public outrage ensued in the Philippines (Larson et al., 2019). Those strongly agreeing that vaccines are safe dropped from 82% to 21% between 2015 and 2018. This demonstrates the particular importance of community education and engagement programmes for dengue vaccines. Successful implementation of vaccination programmes requires robust delivery platforms. A mix of school and clinic-based approaches may be needed to maximise coverage, especially in rural areas (Wilder-Smith et al., 2025).

5 Future Questions

Despite significant advances in dengue control, questions remain surrounding new vaccines, the long-term efficacy of *Wolbachia*-based control and the optimal integration of interventions. The efficacy of TV-003/TV-005 against VCD due to DENV-3 and DENV-4 is inconclusive due to low circulation during the first two years of phase 3 trials in Brazil, with the results for the remainder of the trial not yet released (Lee et al., 2025). There are concerns about the evolutionary stability of the *Wolbachia*-*Ae. aegypti*-DENV interaction (Edenborough et al., 2021). *Wolbachia* imposes a selective pressure favouring any DENV strain which has higher compatibility with *Wolbachia*-infected mosquitoes. It is, therefore, uncertain whether *Wolbachia* fixation will remain an effective dengue control strategy. Super-infection has been proposed as a method to combat reductions in the efficacy of single *Wolbachia* strains (Dorigatti et al., 2018). While vaccination and *Wolbachia* release strategies have been modelled extensively in isolation, there is limited published work investigating their integration (Ndii, 2020; Ndii et al., 2024). Additionally, limited temporal data constrains long-term modelling, while poor surveillance, under-reporting, and misdiagnoses lead to underestimates of the true burden of disease (Cattarino et al., 2020).

6 Project: Aims and Experimental Approach

The integration of vaccination programmes and *Wolbachia* release is likely to be central to an optimal solution for dengue control, although research into this is limited. Considering this, the primary aim of the project is the quantification of the potential impacts of routine vaccination with TAK-003 and *Wolbachia* releases on the dengue burden in Indonesia at a sub-national level, with investigation into the optimal combination of the two interventions. The project secondarily aims to develop maps highlighting operationally feasible target areas for vaccination and *Wolbachia* release. The project will use an age-structured SIR-style framework where the population is divided into susceptible, infected, cross-protected and recovered compartments. The number of times an individual has been infected will be tracked and stochastic simulations will be used to account for the inherent randomness of disease spread. The model will be used to simulate epidemic trajectories under different counterfactual scenarios and assess the potential impacts of different interventions.

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